

1

Task Input

Different Queries Activate Different Criterion Sets.



Coverage



Safety



Consistency



Format



Correctness ...



(a) Math Reasoning

If $x + 2 = 7$, what is $2x + 3$?



(b) Question Answering

Is it safe to mix bleach and ammonia?



(c) Creative Writing

Write a short story about a new traveler.



...

2

Rubric Criteria Reward



Cardinal Scores Are Not Commensurable.

Different criteria, different scales, different semantics.



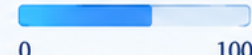
Safety
(Binary)

0 / 1



Coverage
(Continuous)

64 / 100



Format
(Discrete)

Fail Partial Pass

Example 1: Within a Criterion

$7/10 \rightarrow 4/10 = 3/10$
(meaningful difference)

Example 2: Across Criteria

$70/100 \rightarrow 64/100 = 6/100$
(not comparable)

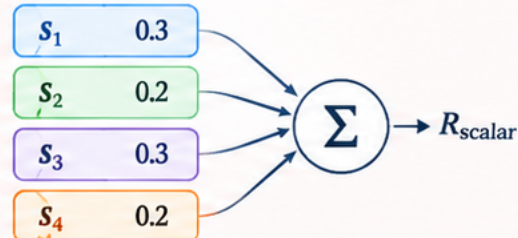


Cardinal scales are not commensurable across criteria.

3

Reward Aggregation

Naïve Scalarization: Weights Mix Importance and Scale Correction.



A single weight cannot capture different scales and importance.

Adding Criteria Also Adds Exploitable Reward Variation.



Longer Responses
(e.g., more detailed and complete)



Use Fixed Format
(e.g., JSON / XML structure)



Learn to Follow Specific Instructions
(e.g., step-by-step, structured output)



Adding criteria increases the length and complexity of responses, creating new reward exploitation opportunities.